EXPERIMENT 21

Create a python program that performs syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

PROGRAM:

```
import nltk

from nltk.tokenize import word_tokenize

from nltk import pos_tag, RegexpParser

from nltk.corpus import wordnet

# Download resources (run once)

nltk.download('punkt')

nltk.download('averaged_perceptron_tagger')

nltk.download('wordnet')

nltk.download('averaged_perceptron_tagger_eng') # Added this line

sentence = input("Enter a sentence: ")

words = word_tokenize(sentence)

tags = pos_tag(words)

grammar = "NP: {<DT>?<JJ>*<NN.*>+}"

parser = RegexpParser(grammar)

tree = parser.parse(tags)

print("\nNoun Phrases and Meanings:\n")

for subtree in tree.subtrees():

  if subtree.label() == "NP":

    phrase = " ".join(word for word, tag in subtree.leaves())

    print("Noun Phrase:", phrase)

    synsets = wordnet.synsets(phrase)

    if synsets:

      print("Meaning:", synsets[0].definition())

    else:
```

```
        print("Meaning: Not found")
    print()
```

OUTPUT:

```
[nltk_data]     /root/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
Enter a sentence: The intelligent student reads a science book.

Noun Phrases and Meanings:

Noun Phrase: The intelligent student
Meaning: Not found

Noun Phrase: a science book
Meaning: Not found
```